

MODULE 10 · DOMAIN 4 - DESIGN COST-OPTIMIZED ARCHITECTURES

Compute Cost and Purchasing Models

Spot, Reserved, Savings Plans and right-sizing.

DOMAIN 4 of 4 EXAM WEIGHT 20% MODULE 10 of 22

AWS discounts **commitment** and **flexibility**.
Promise to keep spending, or promise to
give capacity back, and you pay less.

The purchasing models

MODEL	DISCOUNT	CHOOSE IT WHEN
On-Demand	None. The baseline	Short-term, unpredictable, cannot be interrupted
Spot Instances	Up to about 90%	Interruption tolerated: batch, CI, stateless workers, big data
Reserved Instances	Up to about 72%	Steady usage, and you can name the instance family
Savings Plans	Up to about 72%	Steady spend, but you want flexibility across families or services
Dedicated Hosts	Varies	Licensing tied to physical hardware. Not a cost play

ON THE EXAM

'Can tolerate interruption' means Spot. 'Steady predictable usage' means a commitment. Those two phrases decide most cost questions.

Reserved Instances or Savings Plans?

RESERVED INSTANCES

Commit to specific instances

- Tied to instance family and Region
- Standard RIs can be sold in the Marketplace
- Convertible RIs allow exchange for other types
- Can reserve capacity in a specific AZ

SAVINGS PLANS

Commit to hourly spend

- Compute Savings Plans cover EC2, Fargate and Lambda
- Automatically applies across families and Regions
- Simpler to manage at scale
- Cannot be sold or transferred

THE TIEBREAKER

'We do not know what instance types we will need' means Savings Plans. 'We need guaranteed capacity in an AZ' means a zonal Reserved Instance.

Spot, used properly

- ◆ **Two minute interruption notice.** Workloads must checkpoint or be safely restartable.
- ◆ **Ideal for** batch processing, CI builds, big data, stateless web tiers, and containerised workers.
- ◆ **Never for** databases, licence servers, or anything holding state that cannot be rebuilt.
- ◆ **Mixed instances policy** in an Auto Scaling group blends On-Demand baseline with Spot for the surge.
- ◆ **Spot Fleet** draws from several instance pools to reduce the chance of mass interruption.

ON THE EXAM

The safest exam answer for a variable web tier is often On-Demand for the baseline and Spot for the peak, in one Auto Scaling group.

Reducing cost without changing purchasing model

- ◆ **AWS Compute Optimizer** analyses utilisation and recommends right-sizing. The answer to 'how do we find over-provisioned instances'.
- ◆ **Graviton processors** typically offer better price performance for compatible workloads.
- ◆ **Stop non-production environments** outside working hours. Often the largest single saving available.
- ◆ **Move to managed or serverless** so you stop paying for idle capacity entirely.
- ◆ **Reserving instances you do not need** locks in waste for one to three years.

ON THE EXAM

Right-size before you commit. Buying a three-year reservation for an over-provisioned instance is the classic expensive mistake.

A company runs a nightly data processing job on two hundred EC2 instances for about four hours. The job is fully checkpointed and any lost instance simply has its work reassigned. The instances currently run On-Demand and the finance team has asked for the largest possible saving on this workload.

What should the solutions architect recommend?

- ☐ A Purchase three-year Reserved Instances for two hundred instances
- ☐ B Run the job on Spot Instances
- ☐ C Purchase a Compute Savings Plan covering the nightly usage
- ☐ D Move the job to larger instances to reduce the count

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Three-year Reserved Instances

Reservations bill for every hour of the term. Buying 200 for a job that runs four hours a night wastes most of what you pay for.

C Compute Savings Plan

Better than reservations here since it covers spend rather than instances, but it still discounts less than Spot for interruptible work.

B Spot Instances

Checkpointed and restartable work is exactly Spot's use case, and it gives the deepest discount available.

D Fewer larger instances

Roughly the same total compute at roughly the same price. Instance size is not where the saving is.

THE SIGNAL

'Checkpointed', 'restartable' or 'tolerates interruption' means Spot, and Spot beats every commitment discount.

A company runs a steady production workload costing about eighteen thousand dollars a month across EC2, Fargate and Lambda. Usage has been stable for two years and is expected to continue.

The platform team expects to change instance families over the next year as they migrate some services to Graviton, and they may move workloads between two Regions.

Which purchasing option is MOST appropriate?

- | | |
|--|---|
| ☐ A Standard Reserved Instances for the current instance families | ☐ B A Compute Savings Plan |
| ☐ C An EC2 Instance Savings Plan | ☐ D Spot Instances with a mixed instances policy |

◆ PAUSE AND DECIDE

Which purchasing option is MOST appropriate?

- A

Standard Reserved Instances

Locks to specific families and Regions, both of which the team says will change. Standard RIs cannot be converted.
- B

Compute Savings Plan

Covers EC2, Fargate and Lambda, and applies automatically across families and Regions as the estate changes.
- C

EC2 Instance Savings Plan

Cheaper per hour but locks to one instance family in one Region, and it does not cover Fargate or Lambda at all.
- D

Spot

This is steady production traffic. Interruption is not acceptable, and Spot offers no commitment discount for it.

THE SIGNAL

Three services plus changing instance families plus two Regions equals Compute Savings Plan. Flexibility is the whole point.

A development team runs forty EC2 instances for testing. Engineers work from 09:00 to 18:00 on weekdays. The instances currently run continuously.

Management has asked for cost reduction without affecting engineers' ability to work during business hours, and without any long-term financial commitment.

Which approach delivers the LARGEST saving?

- ☐ A Purchase one-year Reserved Instances for the forty instances
- ☐ B Stop the instances outside working hours on a schedule
- ☐ C Move the instances to Spot
- ☐ D Right-size the instances one size smaller

◆ PAUSE AND DECIDE

Which approach delivers the LARGEST saving?

- A

 One-year Reserved Instances
Excluded by 'no long-term commitment', and it would discount hours the team does not need at all.
- B

 Scheduled stop and start
45 working hours out of 168 in a week. Stopping the rest removes roughly three quarters of the compute bill, with no commitment.
- C

 Spot for development
Cheap per hour, but engineers lose environments mid-task when instances are reclaimed, which affects their ability to work.
- D

 Right-size one size down
Saves perhaps half the cost of those instances, and only while they run. It does not address the idle 75% of the week.

THE SIGNAL

Development environments running out of hours are the single most common cost question on this exam. Turn them off.

A public web application runs an Auto Scaling group between eight and sixty instances. Eight instances are needed at all times, and the rest scale with demand. Traffic is unpredictable.

The application is stateless and instances can be replaced at any time without affecting users. The company wants to reduce cost while keeping the service continuously available.

Which approach is MOST cost-effective?

- A Run all instances on Spot to maximise the discount
- B Cover the eight baseline instances with a Savings Plan and use Spot for the additional capacity
- C Purchase Reserved Instances for sixty instances
- D Run all instances On-Demand and rely on Auto Scaling

◆ PAUSE AND DECIDE

Which approach is MOST cost-effective?

- A

 Everything on Spot
A simultaneous reclamation of the pool could take the whole service down. The baseline must be guaranteed.
- B

 Savings Plan baseline plus Spot surge
The always-on eight get a commitment discount, the variable surge gets Spot pricing, and the service stays up if Spot capacity disappears.
- C

 Reserve sixty instances
Pays for sixty instances continuously when the average is far lower. Reserving peak capacity is the classic over-commitment error.
- D

 All On-Demand
Keeps the service available and takes no discount at all on eight instances that run every hour of the year.

THE SIGNAL

Split the workload. Commitment pricing for the steady baseline, Spot for the variable part. This blend is a recurring right answer.

A finance director reports that EC2 spending has grown thirty percent in a year without a matching increase in traffic. The infrastructure team suspects instances were sized generously during migration and never revisited.

There are around nine hundred instances across twelve accounts, and nobody knows which are over-provisioned.

What should the solutions architect recommend FIRST?

- ☐ A Purchase Savings Plans to reduce the rate on current usage
- ☐ B Use AWS Compute Optimizer to identify over-provisioned instances and right-size them
- ☐ C Move all workloads to Spot Instances
- ☐ D Set AWS Budgets alerts to cap future spending

◆ PAUSE AND DECIDE

What should the solutions architect recommend FIRST?

- A

 Buy Savings Plans now
Commits one to three years of spending on instances that are known to be too large. It locks in the waste at a discount.
- B

 Compute Optimizer, then right-size
Uses actual utilisation data to find over-provisioned instances across accounts. Reduce what you use before discounting what you buy.
- C

 Everything to Spot
Production workloads cannot all tolerate interruption, and it still runs instances larger than necessary.
- D

 Budgets alerts
Tells you when spending rises. It reports the problem without reducing anything.

THE SIGNAL

Right-size first, then commit. Buying a commitment for over-provisioned capacity is the most expensive ordering mistake in domain four.

An API receives roughly four hundred requests per day, arriving unpredictably. It currently runs on two t3.medium instances behind a load balancer, running continuously for availability.

Each request completes in under two seconds. The team wants to reduce cost and is willing to change the deployment model.

Which architecture is MOST cost-effective?

- A Reserved Instances for the two existing instances
- B API Gateway with AWS Lambda
- C A single smaller instance without the load balancer
- D Fargate tasks running continuously behind the load balancer

◆ PAUSE AND DECIDE

Which architecture is MOST cost-effective?

- A

 Reserve the existing instances
Discounts the rate but keeps paying for two instances and a load balancer running 24 hours a day to serve 400 short requests.
- B

 API Gateway with Lambda
400 short invocations a day costs a tiny amount, and nothing runs between requests. It also removes the load balancer charge.
- C

 One smaller instance
Halves the cost and removes redundancy, while still paying continuously for a workload that is idle almost all the time.
- D

 Fargate running continuously
Removes host management but keeps paying for always-running tasks and the load balancer.

THE SIGNAL

Very low request volume with short execution is the strongest signal for serverless. Discounting idle capacity is worse than eliminating it.

Module 10 takeaways

- ◆ **Tolerates interruption** means Spot. It beats every commitment discount for that kind of work.
- ◆ **Changing instance families, or Fargate and Lambda too,** means Compute Savings Plans.
- ◆ **Development environments** should be stopped outside working hours. Usually the biggest single saving.
- ◆ **Baseline plus variable** means commitment pricing for the baseline and Spot for the surge.
- ◆ **Right-size before you commit.** Reserving over-provisioned capacity locks in waste.

ON THE EXAM

Next: storage cost, where lifecycle policies and access patterns do most of the work.